



Spectroscopic properties of Tm:Bi₄Ge₃O₁₂ crystals grown by the micro-pulling-down method

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ARTICLE INFO

Keywords:

Tm:Bi₄Ge₃O₁₂

Micro-pulling-down method

Spectroscopic property

ABSTRACT

Tm³⁺-doped bismuth germanate crystals (Tm:BGO) with various doping concentrations of 0.5, 0.75, 1.0, 1.5, and 2.0 at.% were grown by the micro-pulling-down method (μ -PD). The effect of thulium concentration on the growth, structure, absorption spectra, emission spectra and fluorescence lifetime of ³H₄ and ³F₄ state was studied. Under excitation of 792 nm, Tm:BGO crystals exhibit two fluorescence bands corresponding to the ³H₄ → ³F₄ and the ³F₄ → ³H₆ transitions. The fluorescence lifetime of ³H₄ state decreased strongly as the thulium concentration increased, while the lifetime of ³F₄ state was relatively constant for all thulium concentrations.

1. Introduction

Tm³⁺-doped laser materials have attracted much attention due to their potential applications in medical surgery, tele-communication, remote sensing, and pumping sources of optical parametric oscillators [1–3]. During the last decade, different Tm³⁺-doped crystalline host materials have been investigated for their favourable properties to obtain an efficient laser emission. Tm³⁺ ions can be efficiently excited by commercial AlGaAs laser diode around 800 nm (Tm³⁺, ³H₆ → ³H₄ transition) and emission spectral intervals are around 1.5 μ m (³H₄ → ³F₄ transition) [4], 1.9 μ m (³F₄ → ³H₆ transition) [5] and 2.3 μ m (³H₄ → ³H₅ transition) [6]. Laser oscillation of Tm³⁺ has been demonstration in numerous hosts such as LiYF₄ [7,8], BaY₂F₈ [9], CaF₂ [10], MgWO₄ [11], CaGdAlO₄ [12], YAlO₃ [13], LuScO₃ [14], and Y₂O₃ [15].

Bismuth germanate crystal Bi₄Ge₃O₁₂ (BGO) belongs to cubic system with the space group of I-43d [16]. It has been studied as a host for Er³⁺, Eu³⁺, Dy³⁺, Pr³⁺ and Nd³⁺ ions, etc., with applications as optical temperature sensors, scintillators and laser gain media [17–21]. In addition, as a host material, BGO possesses many qualities desirable for laser applications such as excellent physical and chemical properties [22]. In 1979, A.A. Kaminskii et al. reported the absorption and emission properties of Tm³⁺-doped BGO crystal grown by the Czochralski method

[23]. The micro-pulling-down (μ -PD) method was used to grow BGO crystals with high quality and transparency, where the crystals have low thermal strain compared to other growth methods [24–26]. Eu:BGO and Nd:BGO crystals have also been grown by the μ -PD method [27–29]. BGO crystals melt congruently and relatively high rate of evaporation of Bi₂O₃ occurs during melt overheating [30].

In this work, Tm:BGO crystals with different doping concentration were successfully grown by the micro-pulling-down method. Spectral properties of the crystals were studied at room temperature. Achievement of stationary conditions for the growth of uniformly shaped (constant diameter) fiber crystals was considered as final goal and positive outcome of this paper.

2. Experimental

2.1. Crystal growth

Tm:Bi₄Ge₃O₁₂ crystals were grown by the μ -PD method [31]. The Tm₂O₃ (99.999%), Bi₂O₃ (99.999%) and GeO₂ (99.999%) powders were used as starting materials and weighed according to the formula Tm_{4x}Bi_{4(1-x)}Ge₃O₁₂ (x = 0.005, 0.0075, 0.01, 0.015, 0.02). After thoroughly mixed in an agate mortar, they were pressed into bulks and sintered in

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the air at 750°C for 24h. The polycrystalline Tm:BGO was melted in a platinum crucible, and then the melt was pulled down through a capillary channel at the bottom of the crucible. The Tm:BGO crystals were grown with constant pulling rate of 0.3 mm/min by pulling down a <210> oriented BGO seed in the air atmosphere. Whatever the starting composition prepared in this work ($\text{Tm} \leq 2$ at.%), we had not seen any evolution of growth Kinetic. As shown in Fig. 1(a)–(e), five crack-free Tm:BGO single crystals with dimensions of $\Phi 2 \times 44\text{--}68$ mm³ were obtained and present a light yellow color. The crystals have a rough appearance on the outside, but the inner part of the crystals was transparent and free of visible defects. It is important to note that, evaporation of the bismuth oxide will happen during the crystal growth, once the melt becomes Bi-poor there is no way to establish constant properties (solidification temperature and following shape of the meniscus) of the melt.

2.2. X-ray powder diffraction

In order to investigate the structure of Tm:BGO crystals, small section of the samples were cut from the as-grown crystals and grounded into powder in an agate mortar. The X-ray powder diffraction (XRD) patterns were recorded by an X-ray diffractometer (Bruker-D2, Germany) with a Cu K α radiation ($\lambda = 0.15406$ nm). The experiments were carried out at a scan width of 0.05° within $2\theta = 15\text{--}90^\circ$.

2.3. Spectra measurements

For spectroscopic measurements, the Tm:BGO samples were cut from the as-grown crystals and then optically polished to flat and parallel faces with the size of $\Phi 2 \times 2$ mm³. The absorption spectra were measured from 400 to 2000 nm by a spectrophotometer (Cary5000, UV-VIS-NIR) at room temperature. The fluorescence spectrum as well as the decay time were recorded by a fluorescence spectrometer (Edinburg

Instrument, FLS1000, U.K.) under 792 nm excitation. All the measurements were carried out at room temperature.

3. Results and discussion

3.1. Crystal structure

The room temperature XRD patterns of Tm:BGO crystals are shown in Fig. 2. The crystal structure of Tm:BGO belongs to the cubic space group I-43d. The diffraction peaks of Tm:BGO crystals are in agreement with JCPDS 84-0505 of Bi₄Ge₃O₁₂. No additional impurity phases were found in the patterns. Due to the different doping concentration of Tm³⁺ ions, the relative intensity of each diffraction peak is different and the diffraction peaks of the XRD patterns were slightly offset to the right compared to the JCPDS 84-0505 of BGO. The lattice constants of Tm:BGO crystals were calculated by JADE software 6.0, using a cubic (I-43d) model and the results are listed in Table 1. The calculation error was smaller than 0.0001 Å. With the increase of doping concentration x , the calculated lattice constant decreased, which is attributed to the smaller radius of Tm³⁺ ion (0.869 Å) compared with that of Bi³⁺ ion (0.96 Å) [32]. The results reveal that a small amount of Tm³⁺ ions did not change the structure of BGO crystal.

3.2. Absorption spectra

The room temperature absorption spectra of BGO crystals doped with 0.5, 0.75, 1.0, 1.5, and 2.0 at.% of Tm³⁺ ions in the wavelength region of 400–2000 nm are displayed in Fig. 3. We note that all the samples have the same absorption spectra. Five absorption bands are located at approximately 475, 680, 791, 1210 and 1738 nm, which are attributed to the transitions from the ³H₆ ground state to ¹G₄, ³F₂+³F₃, ³H₄, ³H₅ and ³F₄ excited states, respectively [34]. Among five main peaks, the peak at 791 nm (³H₆ → ³H₄) matches well with the emission of the commercial AlGaAs LD pumping. As the doping concentration increases from 0.5 at.% to 2.0 at.%, the absorption coefficient at 791 nm increases linearly from 0.32 cm^{−1} to 1.45 cm^{−1} as shown in the inset of Fig. 3. For 2.0 at.% Tm:BGO crystal, the full width at half-maximum (FWHM) of the absorption band at 791 nm is about 2.9 nm. The value is smaller than that of Tm:YAG (4 nm at 780 nm [35]), but comparable

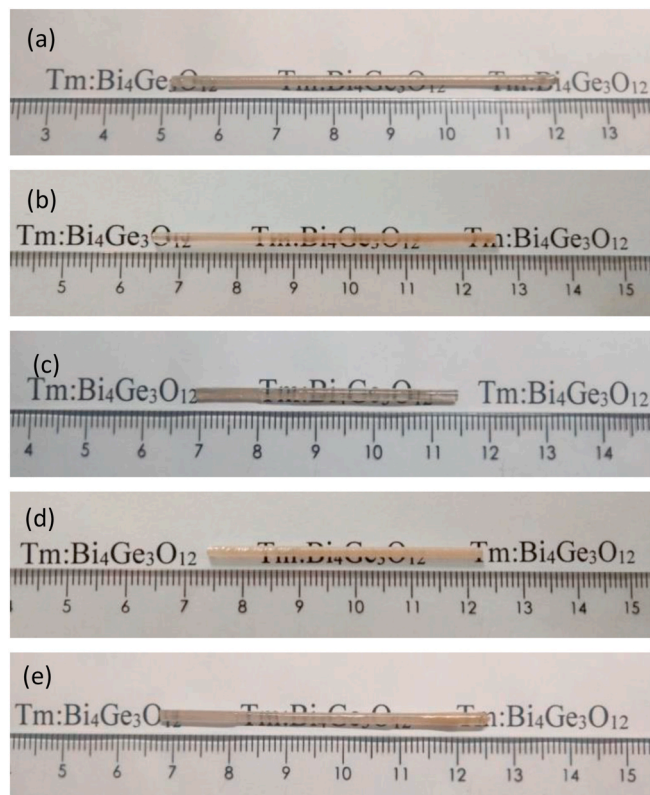


Fig. 1. Photographs of the as-grown Tm:BGO crystals, (a) 0.5 at.%, (b) 0.75 at.%, (c) 1.0 at.%, (d) 1.5 at.% and (e) 2.0 at.%.

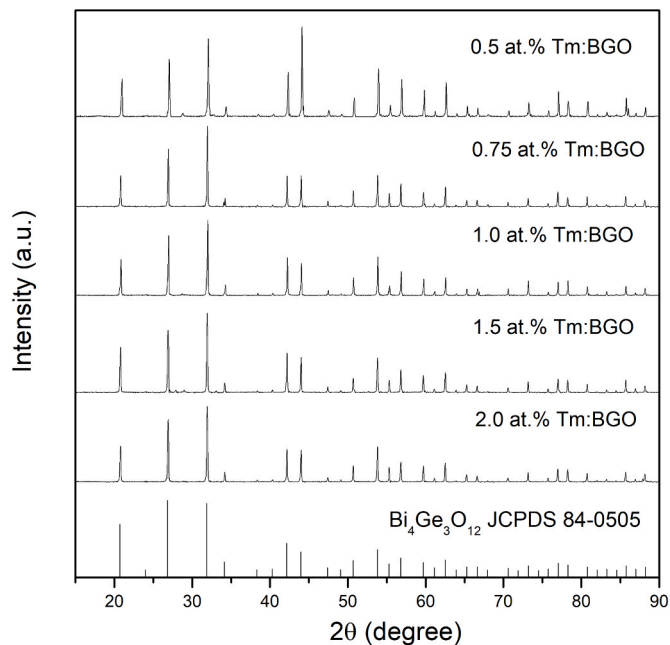


Fig. 2. XRD patterns of Tm:BGO crystals with a standard reference of BGO (JCPD 84-0505).

Table 1

Lattice parameters of Tm:BGO crystals as a function of Tm³⁺ concentration.

x at.% Tm:BGO	a (Å)
0	10.497 [33]
0.5	10.453
0.75	10.451
1.0	10.445
1.5	10.428
2.0	10.419

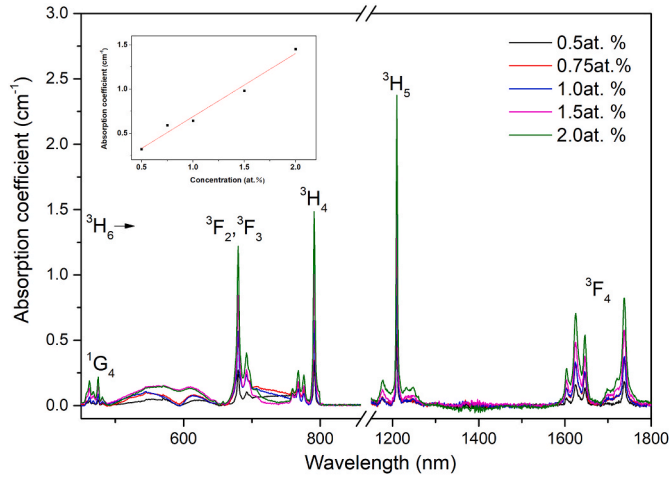


Fig. 3. Absorption spectra of Tm:BGO crystals with different Tm³⁺ concentration.

with that of Tm:KY₃F₁₀ (2.8 nm at 778.6 nm [36]), which is suitable for LD pumping.

3.3. Fluorescence spectra

Under excitation at 792 nm, two fluorescence bands corresponding to the $^3\text{H}_4 \rightarrow ^3\text{F}_4$ and the $^3\text{F}_4 \rightarrow ^3\text{H}_6$ transitions within the measuring range are shown in Fig. 4. At thulium concentrations from 0.5 at.% to 2 at.%, no distinct change in the shape and position of the emission bands was observed in Tm:BGO crystals. We found that the intensity of emissions of the $^3\text{F}_4 \rightarrow ^3\text{H}_6$ transition increases with the increase of thulium concentration. The upper level $^3\text{F}_4$ can be populated by a cross-relaxation process ($^3\text{H}_4 + ^3\text{H}_6 \rightarrow ^3\text{F}_4 + ^3\text{F}_4$) and the effectivity of this process increases with higher thulium concentrations. For 2.0 at.% Tm:BGO crystal, the FWHM of the emission band at 1452 nm was calculated to be 3.9 nm and the FWHM of the emission band at 1879 nm is about 44.6 nm. The FWHM of $^3\text{F}_4 \rightarrow ^3\text{H}_6$ transition is larger than the value of Tm:YAG (31.37 nm [37]). We did not register the emission around 2.3 μm . The energy gap between $^3\text{H}_4$ and $^3\text{H}_5$ energy level is about 4100 cm^{-1} and the Tm³⁺ ions at the $^3\text{H}_4$ level can relax nonradiatively to the $^3\text{H}_5$ level due to the larger phonon energy ($\approx 900 \text{ cm}^{-1}$) of BGO crystal [21].

3.4. Fluorescence lifetime

The fluorescence decay curves of Tm:BGO crystals in correspondence with the $^3\text{H}_4 \rightarrow ^3\text{F}_4$ transition at 1452 nm were measured under 792 nm excitation and are plotted in a semi-logarithmic coordinate system [38, 39] as shown in Fig. 5(a). The decay curves could be fitted with a single exponential decay function well. The fluorescence lifetime was calculated to be 364 μs , 333 μs , 327 μs , 281 μs and 235 μs for the doping concentration of 0.5, 0.75, 1.0, 1.5 and 2.0 at.%, respectively. The lifetime of the $^3\text{H}_4$ state decreased strongly as the thulium concentration

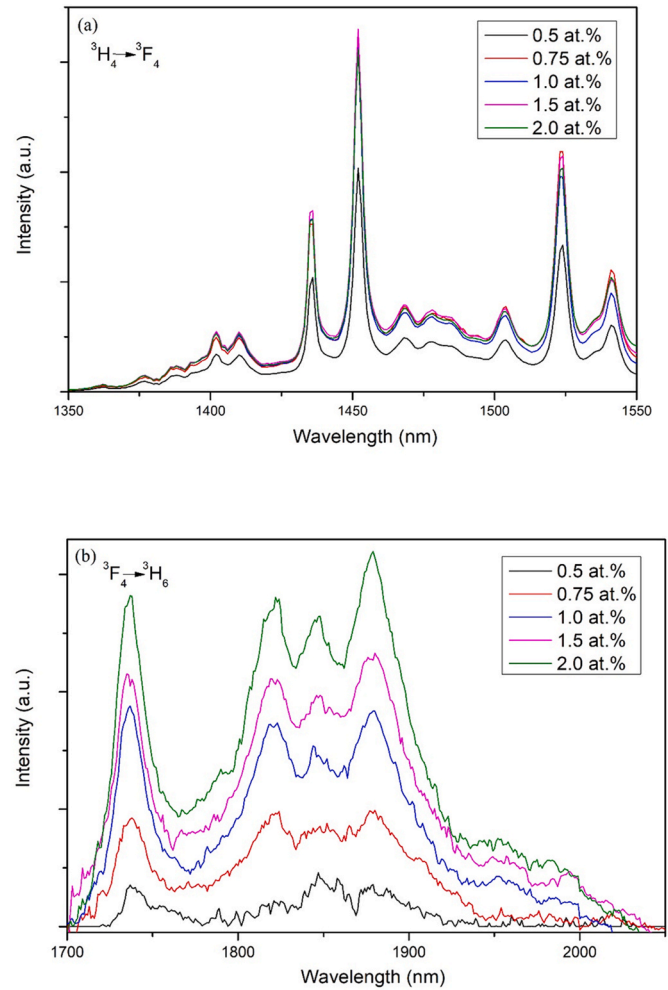


Fig. 4. Fluorescence spectra of Tm:BGO crystals with different Tm³⁺ concentration in (a) 1350–1550 nm and (b) 1700–2150 nm.

increased, which is due to the concentration quenching effect caused by the energy transfer between thulium ions at higher thulium concentration [40]. The fluorescence lifetime at 1452 nm of 1 at.% Tm:BGO is much longer than that of 1% Tm:KGd(WO₄)₂ (175 μs [40]) and 1% Tm:YVO₄ (100 μs [41]).

The fluorescence decay curves of Tm:BGO crystals corresponding to the $^3\text{F}_4 \rightarrow ^3\text{H}_6$ transition at 1879 nm are shown in Fig. 5(b). The measured decay curves showed a single exponential decaying behavior, and the fluorescence lifetime was measured to be 5.22, 5.02, 5.46, 5.28 and 5.38 ms, respectively. In contrast to the lifetime of the $^3\text{H}_4$ state, the fluorescence lifetimes of $^3\text{F}_4$ state are relatively constant for all thulium concentrations. This can be explained by two different mechanisms: the reabsorption effect between $^3\text{F}_4 \rightarrow ^3\text{H}_6$ resonant transitions and repopulation of the $^3\text{H}_4$ state via the $^3\text{H}_4 + ^3\text{H}_6 \rightarrow ^3\text{F}_4 + ^3\text{F}_4$ cross-relaxation process [40]. The fluorescence lifetime at 1879 nm of 1 at.% Tm:BGO is much longer than that of 1% Tm:KGd(WO₄)₂ (1.71 ms [40]) and 1% Tm:YVO₄ (1.03 ms [41]). The results indicated that Tm:BGO single crystal may be a promising laser gain medium.

For 1.45 μm emission, the lifetime of the terminal $^3\text{F}_4$ level is much longer than that of the initial $^3\text{H}_4$ level, making the $^3\text{H}_4 \rightarrow ^3\text{F}_4$ transition self-terminating. The co-doping sensitive ions like Ho³⁺ and Tb³⁺ ions were proposed to depopulate $^3\text{F}_4$ state efficiently [42,43], which will be our next investigation in the near future.

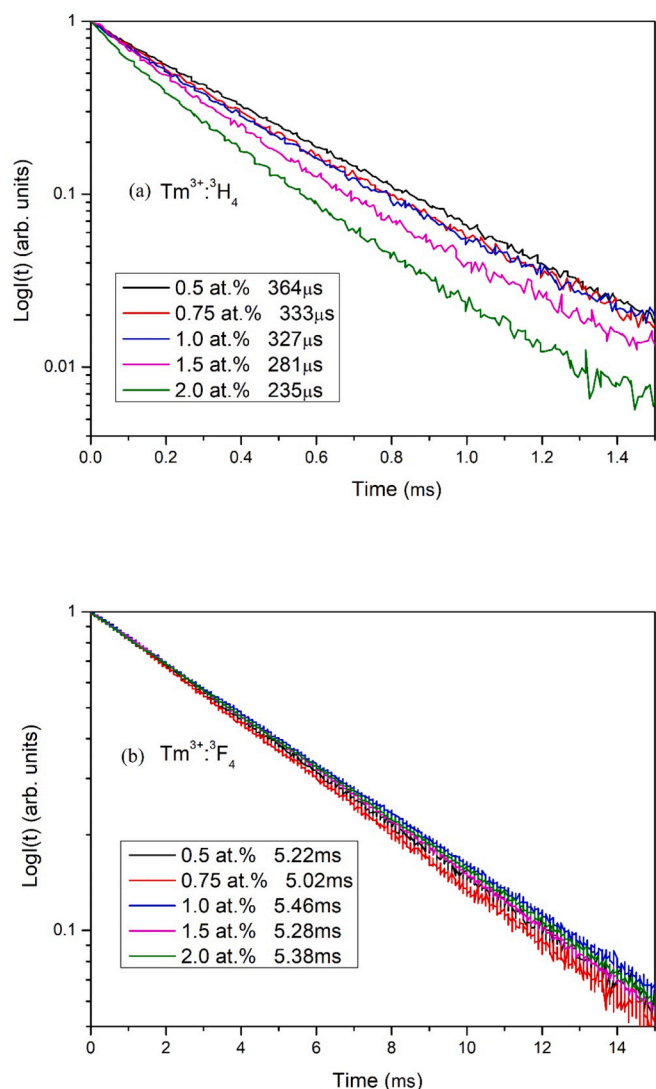


Fig. 5. Fluorescence decay curves of the $^3\text{H}_4$ and $^3\text{F}_4$ level of Tm:BGO crystals.

4. Conclusion

Tm $^{3+}$ -doped BGO crystals with different doping concentration were successfully grown by the μ -PD method. The room temperature XRD revealed that the cell parameter decreased with the increase of doping concentration. The absorption spectra, emission spectra and fluorescence lifetimes of Tm:BGO crystals were measured at room temperature. The absorption peak at 791 nm matches well with the emission of the commercial AlGaAs LD pumping. The intensity of emissions of the $^3\text{F}_4 \rightarrow ^3\text{H}_6$ transition increases with the increase of thulium concentration. The fluorescence lifetime of $^3\text{H}_4$ state decreased strongly as the thulium concentration increased, while the lifetime of $^3\text{F}_4$ state was relatively constant for all thulium concentrations. The results indicate that Tm:BGO crystal could be a promising laser gain medium.

Author contributions

Wen Lu: Crystal growth, Data collection, Writing – original draft, **Jie Xu:** Crystal growth, **Qingsong Song:** Absorption experiment, **Kaiting Bian:** Crystal Growth, **Jun Guo,** XRD experiment, **Jian Liu:** Crystal growth, **Dongzhen Li:** Fluorescence experiment, **Peng Liu:** Data curation, **Chaojin Zhang:** Writing – review & editing, **Xiaodong Xu:** Resources, Writing – review & editing, Supervision, Data curation **Jun Xu:**

Resources, Writing – review & editing, Supervision, Data curation, **Kheirredine Lebbou:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is partially supported by National Natural Science Foundation of China (No. 61621001) and “Qinglan Project” of the Young and Middle-aged Academic Leader of Jiangsu College and University.

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